

RFP-MaaS&Agent Platform-20251111

Model as a Service (MaaS) platform can leverage private computing power resources, enabling enterprises to quickly and efficiently build, deploy, and invoke models without the need to develop and maintain underlying foundational capabilities, significantly reducing the costs of using and managing large models.

- **Plug and Play AI Models:** The MaaS platform provides pre-trained models, allowing users to directly use them through simple API calls without having to train models from scratch, significantly shortening the integration cycle of AI capabilities.
- **Lowering the technical threshold:** The platform encapsulates complex underlying technologies, enabling non-professional developers to quickly and efficiently build, deploy, monitor, and call models, thereby accelerating the implementation of AI in business scenarios.
- **Controllability and Data Security:** The MaaS platform supports adaptation to mainstream open-source large models and mainstream GPU computing power.

1. Model as a Service Platform

1.1 Model Management

1. The Model Plaza supports model classification retrieval. Retrieval methods include: model name, scenario, modality, tag, etc.;
2. Supports viewing detailed information of model cards, including model description, capabilities, usage scenarios, etc.;
3. Supports management operations such as creation, editing, and deletion of model series;
4. Supports adding, editing, and deleting information such as model descriptions, labels, input and output formats. Supports viewing, filtering, and managing model versions;
5. Supports management operations such as creation, editing, deletion, listing/delisting of model entities;
6. Supports manual import of fine-tuned model entities into the platform;

1.2 Model Service

1. Supports model quantization and compression, converting the weights or activation values of the original data source into smaller data types to reduce the requirements for storage space and memory bandwidth;

2. Supports deploying model services through methods such as model weight mirroring, and supports instance management, volume expansion and contraction, and instance release;
3. Supports the targeted scheduling of inference instances to different resource pools or resource groups for deployment, achieving deployment isolation and meeting the deployment requirements of different organizations;
4. Supports monitoring and visual display of the running status of model instances;
5. Supports anti-fragmentation scheduling of inference instances, avoids the emergence of computing power fragmentation, and improves the utilization rate of computing power;
6. Supports providing model services through API interfaces, and can monitor the performance and abnormal conditions of model service operation;
7. Supports creating and managing API Keys for model service calls. Supports API Keys for configuring QPM and TPM for model call flow control;
8. Supports automatic volume expansion and contraction of model instances, and can configure rules such as QPM/TPM thresholds for automatic operation;
9. Supports docking with any third-party language model service that complies with the standard OpenAI protocol, and can be centrally managed through the model gateway;
10. Supports automatic routing to short/long context models based on the requested token length, improving inference performance and reducing resource costs;
11. Supports model service usage statistics, including token statistics, QPS, etc.;
12. Supports the management of routing policies for model services, including creating, modifying, and deleting routing policies;
13. Supports the training datasets for managing language models and multi-modality models, as well as the verification of dataset formats and content;
14. Supports multipart upload of Big data sets exceeding 15GB to relieve bandwidth pressure;

1.3 Model Evaluation

1. Supports creating task evaluation language models, including operations such as task creation, viewing, stopping, and deletion;
2. Supports creating task evaluations for multi-modality models: CogVLM model for image-text understanding;
3. Supports evaluating the effectiveness of the baseline model through public benchmark datasets (such as MMLU, C-Eval, GSM8K, MMMU, OCRBench) and other baseline evaluation datasets;
4. Supports uploading custom datasets for evaluation, which uses the QA comparison mode;

5. Supports automatic generation of test reports, visual viewing of reports, and provides export of test results;
6. Supports result comparison of multiple models or multiple versions on the same test dataset, facilitating the selection of the optimal model;
7. Supports management operations such as creation, editing, and deletion of evaluation datasets;
8. Supports management operations such as importing, viewing, and deleting pre-set evaluation sets;

1.4 Resource Management

1. Supports multi-heterogeneous computing power cluster and node management, and supports global unified management and monitoring of cluster computing power nodes, including node basic information, node resource utilization rate, GPU basic information and utilization rate, etc.;
2. Supports the management of computing power clusters and nodes, viewing cluster nodes, and visualization of cluster resources;
3. Supports viewing cluster monitoring information, including cluster status, resource usage, etc.;
4. Supports management operations such as viewing GPU templates, binding/unbinding nodes, etc.;
5. Supports users to dynamically create resource pools, within which resource groups can be dynamically created according to the requirements of different business scenarios;
6. Supports volume expansion and contraction adjustments at the computing power node level for resource groups, and dynamically monitors the usage of resource groups;
7. Supports scheduling training and inference tasks within a specified resource group, allocating required GPU resources, and configuring scheduling policies;

1.5 Statistical Monitoring

1. Supports audit functionality, visually records and queries information on operations such as data and model access, editing, deletion, etc.;
2. Supports the export function of operational audit events;
3. Supports viewing usage statistics, including model usage, resource usage, etc.;
4. Supports the export function of usage statistics information;
5. Supports operations such as creating, viewing, exporting, and deleting log mining tasks;

1.6 Platform Management

1. Supports basic information management such as creating user accounts and user definitions. Supports creating spaces for different departments and managing space members;
2. Supports User Persona definition and permission assignment to achieve business configuration capabilities. Grants different accounts access permissions to different models;
3. Supports integration with enterprise unified identity authentication and SSO single sign-on;
4. Supports audit functionality, visually records and queries information on operations such as data and model access, editing, deletion, etc.;
5. Supports mainstream training and inference chips;

2. Agent Platform

2.1 Platform Management and Operation

2.1.1 User Management

1. Supports permission division, restriction, and isolation from multiple dimensions such as page menu permissions, operation usage permissions, and data permissions;
2. Supports dynamically adjusting the permission scope of roles when new business resources or scenarios are added to the system;
3. Supports an integrated permission management system, which can cover the permission list of actual business scenarios, supports role permission inheritance, and supports operations such as batch permission changes and revocation for users;
4. Supports the addition of level divisions for the same role and constraints on specific permissions;

2.1.2 Data Management

2.1.2.1 Storage Management

1. Supports the import of Structured Data and unstructured data in different modalities such as text, images, and videos, and supports the connection of mainstream databases;
2. Supports capabilities such as updating, iterating, backtracking, and comparing model dataset versions;
3. Supports configuring sensitive fields and automated desensitization processing of data;

4. Supports the classified and hierarchical storage of data, and grants corresponding access rights to users with different permissions;

2.1.2.2 Data Backflow

1. Supports capabilities such as collection and storage of return data, and whether it supports traceability of return data;
2. Supports monitoring of sensitive data in the return flow and automated filtering and removal of sensitive data;
3. Supports viewing data backflow logs and log analysis capabilities;

2.1.3 Platform Management

2.1.3.1 Log Management

1. Supports recording, deleting, querying, etc. of the development logs of agents, and supports the complete retention and storage of logs;
2. Supports recording user conversation feedback and usage feedback;
3. Supports analysis and processing of agent conversation logs and call logs, and supports automated generation of corresponding analysis reports;

2.1.3.2 Security Policy

1. Supports controlling access restrictions in functions, encrypting sensitive information in backend data storage, etc.;
2. Supports the management backend to modify the configuration of relevant applications, resources, and users at any time.

2.2 Agent Management and Development

2.2.1 Agent Development

2.2.1.1 Custom Template

1. Supports creating agent editing templates based on scenarios and user requests, and allows users to select system prefabricated agents and modify them;
2. Supports secondary editing of saved agent templates;
3. Supports exporting the already built agent templates while retaining the corresponding modules;
4. Does it support importing local template files and online templates, and reusing the intelligent agent's functional modules and configuration parameters with one click?

5. Supports the ability to publish and share templates, and supports functional review of templates to be published;

2.2.1.2 Development Approach

1. Supports no-code development, enabling the main functions of agents to be implemented through user-defined large model prompts, including but not limited to allowing users to define information such as the agent's name, icon, guiding text, tone, and conversation requirements via the configuration page, with the system automatically generating the agent;
2. Supports the integration of foundational large models, prompt templates, and toolchains through methods such as dragging and dropping, thereby completing the functional development of agents;
3. Supports providing complete APIs or SDKs, enabling users to develop the main functions of agents through mainstream programming languages;
4. Supports providing comprehensive development documentation, including but not limited to: introductory guides, detailed descriptions of the development process, usage cases in text-and-image or video format, etc.;
5. Supports providing the preview capability of agent functions, enabling developers to experience the effects of agents before the official release, and supports preview debugging and display of intermediate step results;
6. Supports customizing or uploading effectiveness evaluation datasets, evaluating the effectiveness of agents, and analyzing evaluation results;

2.2.1.3 Capability Expansion

1. Supports platform built-in skill modules such as WEB search, HTTP, FTP, sending emails, surveys, schedule management, weather query, code generation, etc.;
2. Supports platform users to customize the interfaces of callable functions through configuration files in formats such as JSON, and import custom capability functions;
3. Supports smooth switching between large models of different modalities;
4. Supports code interpreter capabilities;
5. Supports long and short-term memory storage for agents, enabling information storage, session record retrieval, etc. in complex tasks;

2.2.2 Agent Management

2.2.2.1 Version Management

1. Supports capabilities such as version update, version rollback, version permission control, version comparison, version information query, version operation log audit, version deletion, parameter modification, etc. for the created agents;

2.2.2.2 Service Management

1. Supports query and retrieval based on information such as the name, type, description, creation time, update time, creator, and status information of the created intelligent agents;
2. Supports management of the listing and delisting of agents in the public and private markets;
3. Supports selective enabling and disabling of agent services;
4. Supports publishing agent capabilities externally in the form of APIs, etc., and connecting them to other systems;
5. Supports intelligent agent label classification based on elements such as industry, scenario, etc.;
6. Supports information feedback on process execution, violation information, capability effectiveness, etc.;

2.2.3 Work Flow Management

2.2.3.1 Component Management

1. Supports basic component modules such as document processing and data processing;
2. Supports visualization higher order component modules such as data analysis, report generation, and chart generation;
3. Supports model components for other scenarios such as Text-to-Image, ASR, TTS, etc.;
4. Supports component canvas operations, including component addition, component connection, etc.;

2.2.3.2 Variable Management

1. Supports variable inputs of common types such as text, json, markdown, images, videos, arrays, etc.;
2. The Agent Platform supports custom variables;

2.2.4 Knowledge Management

1. Supports importing mainstream document formats such as txt, doc, docx, xlsx, etc., supports batch uploading of documents, and supports importing knowledge from external links and online documents;

2. Supports automatic or custom segmentation of knowledge, and supports in-depth knowledge parsing capabilities;
3. Supports the import of professional knowledge, including explanations of proper nouns, import of calculation formulas, etc., and supports knowledge association capabilities;
4. Supports importing documents in common formats, and capabilities such as retrieval-based Q&A and knowledge tracing based on the document Knowledge Base;
5. Supports user-defined knowledge tags, and supports automatically extracting knowledge features and adding tags.

2.3 API Management and Services

2.3.1 API Service

2.3.1.1 Service Security

1. API services support SSL/TLS encryption, user authentication mechanisms (such as OAuth, JWT, etc.), permission verification processes, encrypted storage of sensitive data, security measures to prevent SQL injection, XSS attacks, etc., as well as capabilities such as log auditing, error handling, security bug scanning and repair, which can prevent data leakage, tampering, and unauthorized access;

2.3.1.2 Service Stability

1. Supports exception handling, resource monitoring, load balance capabilities, etc.

2.3.1.3 Service Compatibility

1. The API service of the Agent Platform supports integration with Google Chrome/Microsoft Edge/Firefox Browser/Safari Browser and mainstream databases

2.3.1.4 Service Usability

1. Supports providing API interfaces in the form of HTTP Restful;
2. Supports providing complete interface documentation, including but not limited to interface name, version number, parameters and types, message body format, authentication method, etc.;
3. Support instances of the calling method for the provided interface, which should be based on mainstream programming languages;

2.3.1.5 Measurement Accuracy

1. Supports API external services with metering materials, including but not limited to billing units, billing algorithms, billing cycles, and metering logs;

